

Exact Phase, Projective, and Metric Atlases for a PT-Symmetric Dimer

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Abstract

For the autonomous balanced-gain/loss dimer $i\dot{\psi} = H_{\gamma,\kappa}\psi$, $\kappa > 0$, we give an all-parameter dynamical classification. We additionally compactify the flow on $\mathbb{CP}^1$, separate vector and ray periods, and prove the sharp positivity boundary for an explicit conserved pseudo-Hermitian metric.

1 Frozen system and exact propagator

The frozen release obstruction identifier is HEN-0281. Let

$$H = H_{\gamma,\kappa} = \begin{pmatrix} i\gamma & \kappa \\ \kappa & -i\gamma \end{pmatrix}, \quad \kappa > 0, \quad \gamma \in \mathbb{R}, \quad i\dot{\psi} = H\psi.$$

Parity is $\mathcal{P} = \sigma_x$ and time reversal is componentwise complex conjugation, so $\mathcal{P}H^*\mathcal{P} = H$. Time is the physical propagation parameter; it is not adjusted to external data. Put $\delta = \kappa^2 - \gamma^2$.

Theorem 1 (Complete three-chamber atlas). *The propagator is*

$$e^{-itH} = \begin{cases} \cos(\omega t)I - i\omega^{-1} \sin(\omega t)H, & \delta > 0, \quad \omega = \sqrt{\delta}, \\ I - itH, & \delta = 0, \\ \cosh(\nu t)I - i\nu^{-1} \sinh(\nu t)H, & \delta < 0, \quad \nu = \sqrt{-\delta}. \end{cases}$$

If $\delta > 0$, a ray having nonzero components in both eigenlines has least projective period π/ω , while its vector representative has least period $2\pi/\omega$. The eigenrays themselves are stationary. If $\delta = 0$, H is nonzero rank-one nilpotent with one eigenline, and every generalized state has linear leading growth. If $\delta < 0$, there are two fixed rays; one attracts every generic forward ray and the other repels it, with exponential rate 2ν in their affine ratio.

Proof. Multiplication gives

$$H^2 = (\kappa^2 - \gamma^2)I = \delta I.$$

The even and odd terms of the exponential series give the three formulae. For $\delta > 0$ the eigenvalues are $\pm i\omega$. In their eigenbasis a generic component ratio is multiplied by $e^{2i\omega t}$, proving the ray period; equality of both vector phases first occurs at $2\pi/\omega$. For $\delta = 0$, nonzero determinant-zero H has rank one and the exponential terminates. For $\delta < 0$ the eigenvalues are $\pm i\nu$, so the evolution multipliers are $e^{\pm\nu t}$ and their ratio proves the final assertion. $\square$

This proof is uniform at the exceptional point: it does not diagonalize there or suppress the generalized eigenvector. It also prevents a common factor of two error between vector and ray periods.

2 Projective flow and conserved metrics

On the affine chart $\psi_1 \neq 0$, set $z = \psi_2/\psi_1$. Direct differentiation gives the Riccati equation

$$\dot{z} = i\kappa(z^2 - 1) - 2\gamma z. \tag{1}$$

The reciprocal chart extends it across infinity to a complete holomorphic vector field on $\mathbb{CP}^1$. The complex quadratic discriminant of its fixed point equation is -4δ ; the matrix characteristic discriminant is $+4\delta$. Thus the two fixed rays merge precisely on $|\gamma| = \kappa$. These opposite signs are conventionally harmless but must not be conflated.

Theorem 2 (Sharp metric boundary). *With*

$$Q = \sigma_x, \quad \eta = I + \frac{\gamma}{\kappa} \sigma_y,$$

one has $H^\dagger Q = QH$ and $H^\dagger \eta = \eta H$. Both quadratic forms are conserved. The form Q is indefinite for every parameter, while η is positive definite exactly when $|\gamma| < \kappa$, singular positive semidefinite at $|\gamma| = \kappa$, and indefinite when $|\gamma| > \kappa$.

Proof. The intertwining identities follow by direct Pauli-matrix multiplication. For either matrix M , differentiation gives

$$\frac{d}{dt}(\psi^\dagger M \psi) = i\psi^\dagger (H^\dagger M - MH)\psi = 0.$$

The eigenvalues of η are $1 + |\gamma|/\kappa$ and $1 - |\gamma|/\kappa$, proving the signature claim. $\square$

The standard Euclidean norm is not substituted for these forms:

$$\frac{d}{dt} \|\psi\|^2 = 2\gamma \psi^\dagger \sigma_z \psi.$$

It is generally not conserved. In the unbroken phase the two real-energy eigenlines have opposite Q -Krein signs; their collision makes the normalized positive metric singular at the exceptional point.

References

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